

## Introduction

We hope to solve the equation

$$i\partial_t\psi = (-i\sigma_x\partial_x + m\sigma_z + f(x))\psi$$

where  $\psi = (\psi_1\psi_2)^T$ , and  $f$  is some potential function. In theory, this is only slightly different from Schrödinger. If we can store the spinor  $\psi$ , nearly all the same methods can be used to solve Dirac; we will use Strang splitting to approximate the solution by solving each term individually. Note that  $\propto$  can usually be interpreted as "equal up to normalization."

## Splitting

Like in Schrödinger, we approximate the solution using splitting of the unitary transformation, but this time we use the Strang splitting:

$$\psi(x, t_{j+1}) = e^{-\partial_x\sigma_x\frac{\Delta t}{2}} e^{-i(m\sigma_z+f(x))\Delta t} e^{-\partial_x\sigma_x\frac{\Delta t}{2}} \psi(x, t_j) + O(\Delta t^3)$$

With  $\frac{t}{\Delta t}$  timesteps, this gives a second order in  $\Delta t$  accumulated error  $O(\frac{t}{\Delta t}\Delta t^3) = O(t\Delta t^2)$ . Also recognize that  $[m\sigma_z, f(x)I] = 0$ , so we can split exactly  $e^{-i(m\sigma_z+f(x))\Delta t} = e^{-im\sigma_z} e^{-if(x)}$ . The implementation of these operators and storage of the spinor on qubits are detailed below.

## Discretization and Storage

Fortunately, the problem of storing the spinor on qubits has a simple solution: we maintain a register to store position, as in Schrödinger, and add one qubit as an indicator. We discretize  $\psi_1$  and  $\psi_2$  into  $|\psi_1\rangle$  and  $|\psi_2\rangle$  as we know how

from Schrödinger, and store the state  $\psi$  as

$$\begin{aligned}
|\psi\rangle &\propto \sum_{k=0}^{N-1} (\psi_1(x_k) |0\rangle + \psi_2(x_k) |1\rangle) \otimes |k\rangle \\
&= |0\rangle \otimes \left( \sum_{k=0}^{N-1} \psi_1(x_k) |k\rangle \right) + |1\rangle \otimes \left( \sum_{k=0}^{N-1} \psi_2(x_k) |k\rangle \right) \\
&\propto |0\rangle \otimes |\psi_1\rangle + |1\rangle \otimes |\psi_2\rangle
\end{aligned}$$

With the indicator as the most significant qubit, this is equivalent to concatenating the column vectors of amplitudes for  $|\psi_1\rangle$  and  $|\psi_2\rangle$ . Regarding measurement probabilities, if we measure only the register storing position, and leave the indicator qubit unmeasured, the probability of measuring  $|k\rangle$  is  $p(k) \propto |\psi_1(x_k, t)|^2 + |\psi_2(x_k, t)|^2$ . This is justified as follows: the state  $|\psi\rangle$  has density matrix

$$\begin{aligned}
|\psi\rangle \langle\psi| &\propto (|0\rangle \otimes |\psi_1\rangle + |1\rangle \otimes |\psi_2\rangle)(\langle 0| \otimes \langle\psi_1| + \langle 1| \otimes \langle\psi_2|) \\
&= |0\rangle \langle 0| \otimes |\psi_1\rangle \langle\psi_1| + |0\rangle \langle 1| \otimes |\psi_1\rangle \langle\psi_2| + |1\rangle \langle 0| \otimes |\psi_2\rangle \langle\psi_1| + |1\rangle \langle 1| \otimes |\psi_2\rangle \langle\psi_2|
\end{aligned}$$

Measurement probabilities of the position register are given by the reduced density matrix obtained by tracing out the spinor qubit. Taking the partial trace, we get

$$\begin{aligned}
\rho &\propto |\psi_1\rangle \langle\psi_1| + |\psi_2\rangle \langle\psi_2| \\
&= \sum_{l=0}^{N-1} \sum_{j=0}^{N-1} (\psi_1(x_l) \psi_1(x_j)^* + \psi_2(x_l) \psi_2(x_j)^*) |l\rangle \langle j|
\end{aligned}$$

Thus the probability of measuring  $|k\rangle$  at time  $t$  is  $p(k) = \text{tr}(|k\rangle \langle k| \rho) \propto |\psi_1(x_k, t)|^2 + |\psi_2(x_k, t)|^2$  by orthonormality of the computational basis.

## Transport Term

We concern ourselves with the equation  $i\partial_t\psi = -i\partial_x\sigma_x\psi$ . Recognizing that  $\sigma_x = H\sigma_zH$ , where,  $H$  is the Hadamard matrix, this system is equivalent to  $\partial_t H\psi = -\partial_x\sigma_z H\psi$ . Letting  $\varphi := (\varphi_1, \varphi_2)^T = H\psi$ , since  $\sigma_z$  is diagonal, the system is decoupled:  $\partial_t\varphi = -\partial_x\sigma_z\varphi$ . Equivalently,

$$\partial_t\varphi_1 = -\partial_x\varphi_1$$

$$\partial_t\varphi_2 = \partial_x\varphi_2$$

This is a first order linear transport equation, which has a known solution for initial conditions  $\varphi_j^0(x) = \varphi_j(x, 0)$ :  $\varphi_1(x, t) = \varphi_1^0(x-t)$  and  $\varphi_2(x, t) = \varphi_2^0(x+t)$ . For the sake of implementation on a quantum computer, we solve using the Fourier transform  $\mathcal{F}$  to get

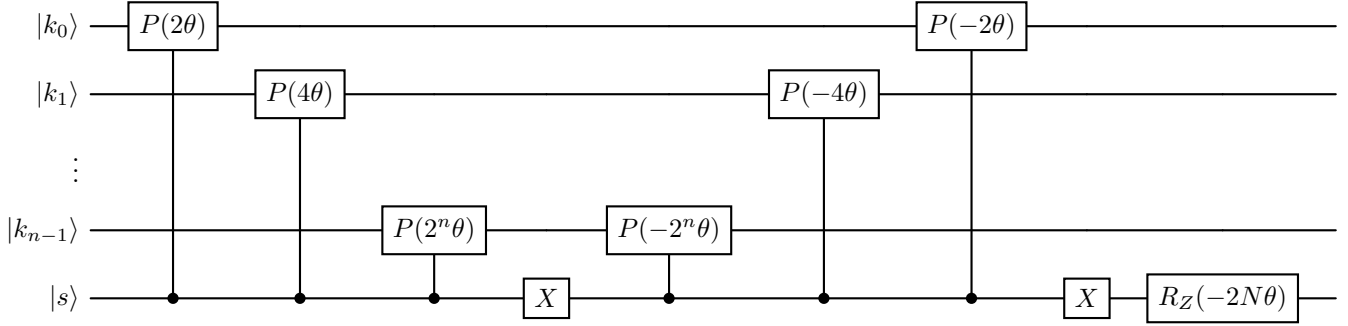
$$\varphi(x, t) = \begin{bmatrix} \mathcal{F}^{-1}(e^{-i\xi t}\mathcal{F}(\varphi_1^0)(\xi)) \\ \mathcal{F}^{-1}(e^{i\xi t}\mathcal{F}(\varphi_2^0)(\xi)) \end{bmatrix}$$

Hence on the quantum computer we map  $|s\rangle \otimes |k\rangle \mapsto |s\rangle \otimes (QFT e^{(-1)^{s+1}i\xi_k\Delta t} QFT^\dagger |k\rangle)$ , where as before,  $\xi_k = \frac{\pi}{L}(2k - N)$ . A new concern we have in Dirac is that any  $\theta$  dependent global phase incurred in the application of  $e^{i\theta k}$  is now a relative phase between  $\varphi_1$  and  $\varphi_2$  that we need to correct for. We write the phase shift operation more carefully with respect to global phase changes. Note that I've changed the implementation of  $e^{ik} := \bigotimes_{j=0}^{n-1} P(2^j)$  to use phase gates, rather than differences of identity and  $Z$  gates. This makes global phase easier to keep track of.

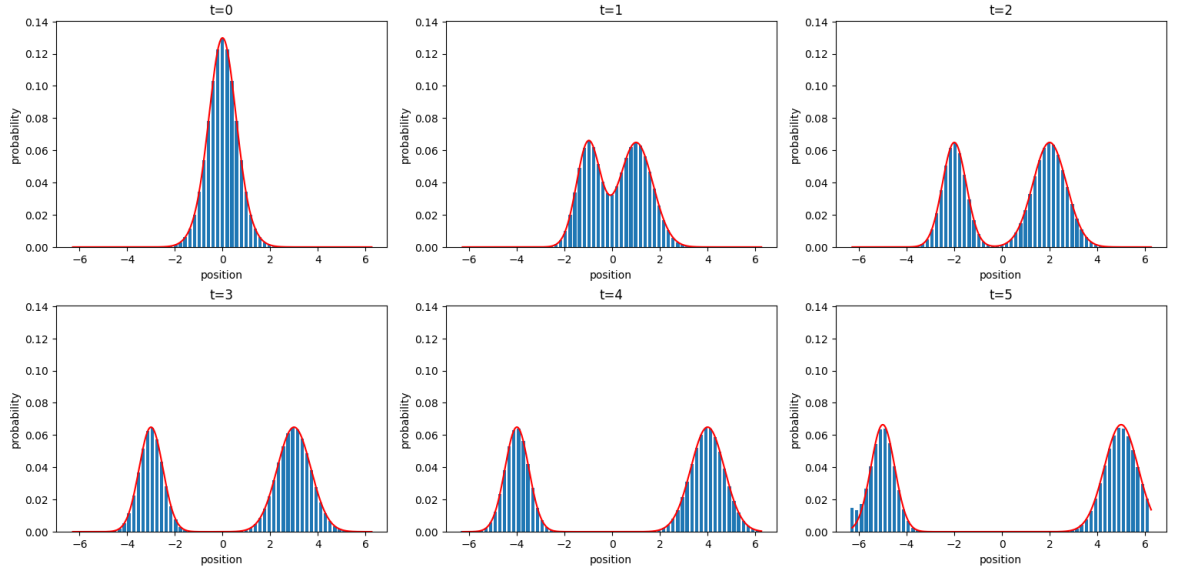
$$\exp\left((-1)^{s+1}i\frac{\pi}{L}(2k - N)\Delta t\right) = \exp\left((-1)^s iN\frac{\pi}{L}\Delta t\right) \exp\left((-1)^{s+1}2i\frac{\pi}{L}\Delta tk\right)$$

$$= \exp\left((-1)^s i N \frac{\pi}{L} \Delta t\right) \bigotimes_{j=0}^{n-1} P\left((-1)^{s+1} 2^{j+1} \frac{\pi}{L} \Delta t\right)$$

$\varphi_1$  picks up a phase of  $e^{\frac{\pi i N \Delta t}{L}}$ , and  $\varphi_2$  picks up a phase of  $e^{-\frac{\pi i N \Delta t}{L}}$ , which we build in by applying  $R_Z\left(\frac{-2\pi N \Delta t}{L}\right)$  to the spinor qubit. The quantum circuit we implement to apply the shift phase is below.  $\theta = \frac{\pi}{L} \Delta t$ .



Sandwiching this with  $QFT$  and  $QFT^\dagger$  solves the linear transport equation, which is certainly interesting on its own, so I have included a plot below of a solution with two Gaussians  $\varphi_1$  and  $\varphi_2$ .



As interesting as this is, our concern is with Dirac, and still we need to be able to map between  $|\psi\rangle$  and  $|\varphi\rangle$ . This is fortunately very simple and reveals itself after a bit of algebra. Recalling that  $\psi = H\varphi$  and  $\varphi = H\psi$ , and we have the state  $|\varphi\rangle \propto |0\rangle |\varphi_1\rangle + |1\rangle |\varphi_2\rangle$

$$\begin{aligned}
|\psi\rangle &\propto |0\rangle |\psi_1\rangle + |1\rangle |\psi_2\rangle \\
&= \frac{|0\rangle}{\sqrt{2}} (|\varphi_1\rangle + |\varphi_2\rangle) + \frac{|1\rangle}{\sqrt{2}} (|\varphi_1\rangle - |\varphi_2\rangle) \\
&= \frac{|0\rangle + |1\rangle}{\sqrt{2}} |\varphi_1\rangle + \frac{|0\rangle - |1\rangle}{\sqrt{2}} |\varphi_2\rangle \\
&= (H \otimes I) |\varphi\rangle
\end{aligned}$$

The same gate maps both directions, so we can transform between  $|\varphi\rangle$  and  $|\psi\rangle$  with a Hadamard gate on the spinor qubit. Thus our solution to  $\partial_t \psi = -\sigma_x \partial_x \psi$  is given by simply applying  $H$  to the spinor qubit, following the work above for the linear transport, then applying another  $H$ . This is the circuit returned by `transport()`.

## Mass Term

We now concern ourselves with the mass term  $i\partial_t \psi = m\sigma_z \psi$ . Fortunately,  $\sigma_z$  is diagonal in the standard basis, so we have the following solution.

$$\psi(x, t) = \begin{bmatrix} e^{-imt} \psi_1^0(x) \\ e^{imt} \psi_2^0(x) \end{bmatrix}$$

On qubits we want to map  $|0\rangle |\psi_1\rangle + |1\rangle |\psi_2\rangle \mapsto e^{-im\Delta t} |0\rangle |\psi_1\rangle + e^{im\Delta t} |1\rangle |\psi_2\rangle$ .

This is implemented with a single  $R_Z(2m\Delta t)$  on the spinor qubit.